



Stock Prediction and Visualization

MINOR PROJECT BY :-

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Introduction

The stock market is a financial environment where shares of companies are traded on a daily basis. Stock prices are constantly changing due to demand and supply, company performance, economic conditions, political events, and global market trends. Because of these continuous changes, predicting stock prices accurately is a complex task. Traditional stock market analysis requires deep financial knowledge, technical indicators, and experience, making it difficult for beginners to participate effectively. This project focuses on simplifying stock market prediction by using Artificial Intelligence and Machine Learning techniques. The system analyzes historical stock price data and identifies patterns that help forecast future price movements. The predicted results are presented using clear tables and simple visuals so that users can understand market trends without confusion.

Problem Statement And Objective

Problem Statement Problem Statement and Objectives Stock prices are highly volatile and influenced by unpredictable factors such as economic changes, company announcements, global events, and investor behavior. Manual stock analysis is time-consuming and often affected by emotional bias, which may result in financial losses. Many existing prediction tools are complex and require advanced financial knowledge, making them unsuitable for beginners.

The objective of this project is to develop an AI-based system capable of predicting stock prices using historical data. The system aims to provide real-time insights, user-friendly visualization, and educational support to improve financial understanding among users

Literature Survey

Stock price prediction has been researched extensively over the years. Early approaches relied on statistical models such as Linear Regression and ARIMA. While these models were easy to implement, they failed to capture complex and non-linear market behavior. Machine learning techniques later improved prediction accuracy by analyzing multiple market variables simultaneously.

Recent research highlights the effectiveness of deep learning models such as Long Short-Term Memory networks in handling time-series data. Reinforcement Learning techniques have also been explored to optimize trading strategies by learning from past market behavior. This project is influenced by these modern research approaches.

Methodology

The project follows the Agile development methodology, which supports iterative development and continuous improvement. Stock market data is collected from reliable sources such as Kaggle and Yahoo Finance. The collected data is cleaned, normalized, and prepared to improve prediction accuracy.

Long Short-Term Memory models are trained using historical stock data to identify trends and predict future prices. Reinforcement Learning techniques further enhance predictions by optimizing strategies based on market feedback. The final results are displayed using simple dashboards and tables.

Expected Outcomes and Conclusion

The proposed system is expected to provide accurate stock price predictions using Artificial Intelligence and Machine Learning techniques. It helps users understand market trends through visual representation and supports data-driven decision-making. The application is designed to be simple, reliable, and accessible while offering meaningful insights for both beginners and experienced investors.

In conclusion, this project demonstrates how AI-based models can improve stock market analysis. Future enhancements may include portfolio management tools, risk analysis modules, and advanced deep learning models to further improve prediction accuracy and usability.